Instructor Guide: Chart Critique

Type: Exercise (pairs or small groups) **Time:** 10 minutes (6 min pair work + 4 min debrief)

Pre-Activity Setup

- Prepare the five-chart handout before class. Use the OrbitCo-style sample data so the numbers look operational, not abstract. The canonical set:
  1. **Chart 1 — Truncated y-axis on uptime columns.** Weekly uptime percentages displayed as a column chart with the y-axis starting at 99.5%. Four weeks, values 99.92 / 99.78 / 99.62 / 99.91. Visually the third week looks catastrophic; it is a 0.3-point variation.
  2. **Chart 2 — 3D pie with twelve slices.** Tickets by resolution code, 3D perspective, shadowed slices. Labels overlap. Legend off to the side.
  3. **Chart 3 — Line chart connecting unordered categorical data.** Ticket count by severity (High, Medium, Low) with the three points connected by a line. Implies a trend where none exists.
  4. **Chart 4 — Well-designed horizontal bar.** Top 8 resolution codes sorted by count, horizontal bars, zero-based axis, clear title ("Top resolution codes, last 30 days"), readable data labels on each bar.
  5. **Chart 5 — Misleading color on a team breakdown.** Four teams shown as column chart with alternating red and green fills (no ordinal meaning). Colorblind-hostile and implies a pass/fail interpretation where none exists.
- Print enough handouts for every pair to have one copy. Digital copies work too, but paper forces the eye across all five charts before the conversation starts.
- Have a hidden backup slide ready with the "fixed" versions of each problem chart to show during debrief. This is the reveal that makes the lesson stick.

Facilitation Steps

1. **0:00 — Launch** (1 min): Distribute the handout. Read the instructions aloud, emphasizing the "at least one chart is well-designed" clue. "Do not invent problems. If a chart looks fine, say so, and defend it." Start the timer.

2. **0:01 — Pair work** (5 min): Circulate between pairs. Do not disclose which chart is the well-designed one — let the class argue it out. If a pair is stuck, ask "what question is this chart trying to answer?" to unstick them.
3. **0:06 — Debrief** (4 min): Assign each pair one chart. Each pair presents for 30 seconds. After each pair, flip to the fixed-version slide and confirm or challenge the proposed fix. The well-designed chart gets its own 30 seconds — ask the assigned pair why it works before revealing your answer.

What to Circulate For

- **Inventing a problem with Chart 4.** The reflex to find a flaw in every chart is strong. Pairs will look for one. The teachable moment is naming the reflex: "Critique exercises do not mean everything is wrong — part of the skill is defending good design." Redirect them to the "why it works" side of the table.
- **Cosmetic fixes to structural problems.** "Make the 3D pie a different color" is not a fix for a 3D pie with twelve slices. Push for the structural change (2D bar chart, sorted, fewer categories).
- **Surface-level fixes to the truncated axis.** "Move the y-axis labels" is not a fix. The fix is either (a) a zero-based y-axis or (b) switch to a line chart with an explicit annotation. Force them to pick one and defend it.
- **Overthinking the categorical-line chart.** Some pairs will argue the line is "fine because the trend is down." The trend is fictitious — ordering High/Medium/Low is arbitrary. A column chart is the honest answer. If they cannot let go of the line, ask: "If I rearranged the severity order to Low/Medium/High, does the trend change direction?"
- **Missing the colorblind angle on Chart 5.** Pairs often call it "cluttered" but miss the accessibility problem. Nudge with: "What does this chart look like to someone who cannot distinguish red from green?"

Common Pitfalls

- **Pair runs over 30 seconds in debrief.** Hold the line. The time pressure forces a clearer argument, and there are five charts to cover in four minutes.

- **Learners attach emotional significance to "their" chart's critique.** If a pair built a 3D pie in a previous class, they may defend it reflexively. Acknowledge and redirect: "Defenders of 3D pies have lost the argument in the industry. The job now is to recognize when someone else hands you one."
- **Debrief gets stuck on Chart 2 (the 3D pie).** It is the most visually offensive chart in the set and debates about it eat the clock. Move on after two minutes even if the discussion is good — the other three problem charts teach different skills and need their time.
- **No one picks Chart 4 as well-designed.** Sometimes the whole class assumes the well-designed chart must be a trick. The instructor's reveal ("no, Chart 4 is actually fine — here is why") is the debrief moment, and it is worth waiting for the class to converge on the wrong answer before revealing.

Debrief Prompts

- **"Which of these problems have you actually seen in a work or school setting? Where?"** Expected answer: truncated axes in financial reporting decks and 3D pies in older business dashboards are the two most frequent sightings. This grounds the exercise in operational reality rather than academic critique.
- **"If the chart author told you they used the 3D pie because their manager asked for it, how do you push back?"** Expected answer: rebuild the chart as a 2D bar and show both versions side-by-side. Let the comparison make the argument. Pushing back on a manager's visual preference with data is more effective than pushing back with principle.
- **"For the well-designed chart — what makes it work? What could be copied to other dashboards?"** Expected answer: zero-based axis, sort by value for readability, horizontal bars for long labels, title that names the question the chart answers. These four habits cover most operational reporting.

Expected Answers

The reference critique table the class should converge on during debrief:

Chart #	Problem (or "well-designed")	Specific fix (or reason it works)
1	Truncated y-axis on a column chart (starts at 99.5%) exaggerates a 0.3-point uptime dip into a visual cliff.	Either switch to a line chart (rate over time tolerates non-zero baseline) or rebuild the column with a zero-based axis; annotate actual percentages on each bar either way.
2	3D pie with twelve slices: perspective distortion makes front slices look larger than equal-area back slices, and twelve slices is too many for a pie under any dimensionality.	Switch to a 2D horizontal bar chart sorted by count. Keep the full category list; bar length is readable where pie slivers are not.
3	Line chart on unordered categorical data (severity) implies a trend where none exists — the order of categories is arbitrary.	Switch to a column chart. Column bars sit on a shared baseline and make the comparison honest without implying connection.
4	Well-designed — horizontal bar sorted by count, 8 categories, zero-based axis, question-in-the-title.	The sort order makes magnitude comparison instant; horizontal bars handle long category labels; the title ("Top resolution codes, last 30 days") names the question the chart answers so the reader knows what to look for.
5	Misleading color — red/green on non-ordinal categorical data implies pass/fail meaning where none exists, and hurts colorblind readers.	Use a neutral categorical palette (blue / orange / teal / gray, for example). Reserve red and green for status (breach vs OK) where the meaning is conventional.

If a pair proposes a different but equally valid fix (for example, "Chart 2 should be a smaller 2D pie with only the top 4 codes plus 'Other'"), accept it. The goal is defensibility of the fix, not uniformity of the answer.